Lab work MA21
Ellipsometry

G. Lerondel T225
K. Nomenyo T108

Work expected:

During this lab work, you will be asked to actively participate by answering question orally and in the spread-sheet. Morever, there will be questions to prepare beforehand. You will be asked to hand out a detailed report which will be evaluated according to the quality of the answers to the questions and the clarity of explanations of the experiments. Each report must be handed out at the end of the lab work.

The objective of this lab work is to illustrate, via different sample (metal and semiconductor substrates, thin films…) the potential of the ellipsometry technique for the index determination (optical properties of a material) and the thickness determination (thin film deposition).

I- Principle

Let us consider an electromagnetic wave linearly polarised at 45° in the polarisation plane (P, S). After reflection on a surface (sample), this wave remains a plane wave if there is no surface roughness and becomes elliptically polarised.

1) Give the expression of the electromagnetic wave in this case, in a basis and draw a detailed scheme of the geometry.

This change of polarisation is due to the difference of the reflection coefficient amplitudes, r_p and r_s depending whether the wave is within or orthogonal with the incidence wave respectively.

If we project in the polarisation plane (P, S), we note that the parameters of the ellipse are directly related to the reflection coefficients (the moduli of r_p and r_s correspond for example at the projection of the ellipse axis on the vectors P and S respectively).

2) Recall the definition of these coefficients (so-called Fresnel coefficients). We will distinguish 2 cases. What is the definition of the Brewster angle?

By analysing the polarisation after reflection (reconstitution of the ellipse) with the use of an optical polarisor that we rotate, we determine the TWO elliptical parameters (ψ, Δ) as well as r_s and r_p.

$$\frac{r_p}{r_s} = \frac{|r_p| e^{i\delta_p}}{|r_s| e^{i\delta_s}} = \tan(\Psi) e^{i\Delta}$$

with $\Delta=\delta_p-\delta_r$ and $\tan(\Psi)=|r_p/r_s|$. $\tan(\Psi)$ represents the change in amplitude with the reflection and Δ is the dephasing. Ψ and Δ are respectively in the ranges from 0 to 360° and from 0 to 180°.

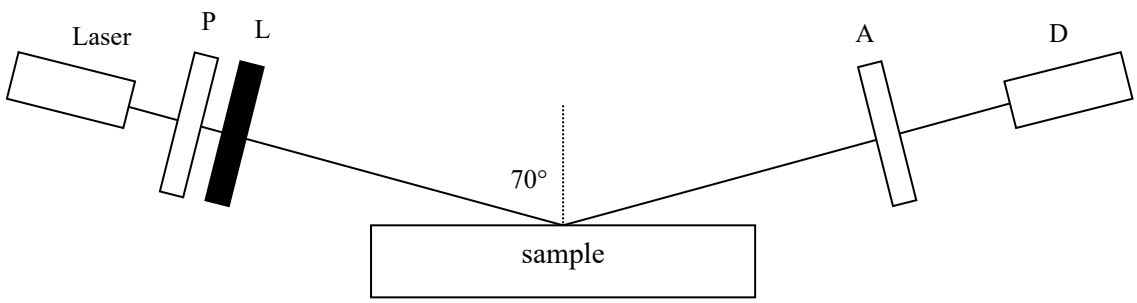
The reflection coefficients are directly related to the optical structure of the sample (substrate, thin film) a model of this structure is then necessary in order to adjust the experimental measurements.

In the « simple » case of a substrate for example (semi-infinite sample) r_p and r_s depend directly and only on the refraction index, with the help of the TWO parameters, the measurement allows us to recover the complex index of the material (refraction + absorption).

3) As an example, give the values of r_p and r_s for an incident beam normal with the interface between air and glass, at the wavelength of 632 nm.

II- Experimental set-up

There are two (in fact three) main types of ellipsometers: the annulations ellipsometer and the rotating birefringent plate ellipsometer. In both cases, the components are the following:



Light is emitted by a linearly polarised laser (after the polarisor P) then go through a quart-wave plate $\lambda/4$ (L) before being reflected on the sample and then analysed with the help of an analysor/polarisor (A) and a detector (D) (silicon photodiode).

The equipment used in this lab work (Ellisel) corresponds to another type of ellipsometer but we will nonetheless describe the principle of the annulations ellipsometer. The actual working mode of the Ellisel is more complicated.

i- Handling of the equipment

Switch on the computer and start the programme DIGISEL**.** We will make sure that in order to "naviguate" in the different options, we use 'Enter' or 'Echap'.

Make sure to use tweezers and gloves (1 personne at a time) to handle the samples. Make sure you do not scratch them.

ii- Calibration

This calibration is made with the help of a silica thin film (SiO_2) deposited on silicon substrate. The optical response of the structure is known:

Substrate index n = 3.8 and κ (imaginary part) = 0.02

Index of the thin film n_{SiO2} = 1.465 and κ=0

We will keep the angle of incidence at 70.9°.

-Start the automatic calibration of the apparatus by setting the parameters above. To start the calibration, we go to 'Calibration' then 'Automatic Calibration' then 'Echap' and the calibration sequence starts. Place the shiny face of the sample towards the top.

- What is thickness measured knowing that we must find a value between 122 and 123 nm. Systematically jot down the values of Psi and Delta. In order to make a measurement, go to 'Development' then select the right configuration.
Note: You are asked for a 'Second layer' that you do not have in principle. In fact, the second layer is air. Why do you think that is? Draw a scheme of the structure.

Data obtained after calibration:

- Thickness e = 124.8 nm
- Ψ = 61,26°
- Δ = 80,30°

- At the end of the calibration, you can test the homogeneity of the film on the entire 3 inch diameter wafer. How can you do that? What is the precision on the control of the thickness? The edges are not as 'clean', why do you think that is?

Data:

Measure N°	1	2	3 (edge)	4 (edge)
e	124,4	125,1	124,8	125,2
Ψ (°)	60,92	62,14	61,26	62,60
Δ (°)	80,93	79,82	80,30	79,48

III- Applications

i- Complex index determination

We will first determine indices of materials. For that, we will look into 3 substrates (semi-infinite media) Sub1, Sub2 and Sub3.
- What do we mean by semi-infinite medium in optics?
- Measure the complex indices of the 3 substrates Sub1, Sub2 and Sub3. Write down systematically the values of Psi and Delta. What can we say about the diffusion of the laseron these two samples?

Data:

	Sub1	Sub2	Sub3
n	3,888	3,839	0,182
k	0,187	0,491	3,2
Ψ (°)	9,25	9,80	43,57
Δ (°)	171,22	157,49	102,85

- Confirm, a posteriori, that we are dealing indeed with « substrates », semi-infinite media and that the initial assumption was justified. For that, you will use Beer-Lambert's law by taking a substrate thickness of 0.5 mm.

- Search what are the materials for these substrates, knowing that Sub1 and Sub2 are semiconductors and Sub3 is a metal? We will use the website https://refractiveindex.info/ and we will look for the material indices: GaP, InAs, Si and ZnO for the semiconductor and we will look for the material indices: gold, iron, nickel and aluminium for the metal.

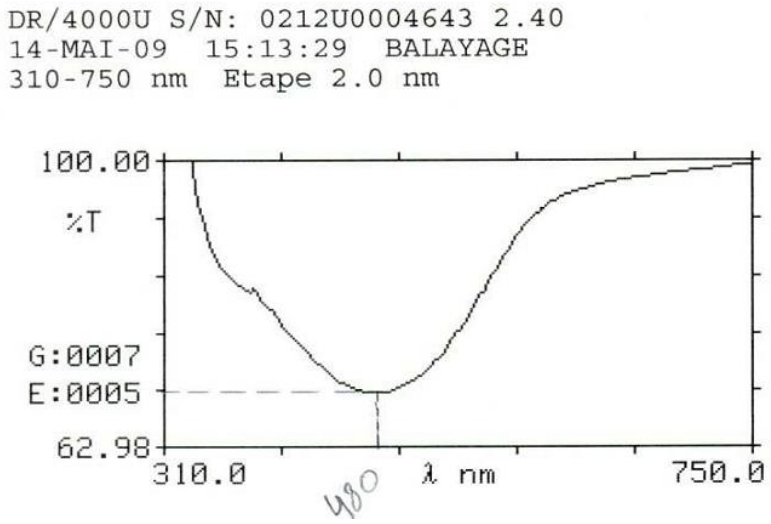
ii- Thickness determination

You have several thin films layers: a layer of DR1-doped PMMA on silicon (3 layers of 3 thicknesses Si1, Si2, and Si3).

-How many parameters are measurable by ellipsometry? Recall them!

-With the help of a simple ellipsometric measurement, can we determine all the parameters of the layer? If yes, in what spectral range knowing that PMMA-DR1 is an absorbing material (cf. lab work on optical transmission). Is it the case at 632.8 nm?

We provide the absorption spectrum of PMMA-DR1:



- Measure the different « measurable » optical parameters for the thin films of DR1-doped PMMA on silicon (Si1, Si2 and Si3). We will make sure to detail the measurement method knowing that the layer is deposited on substrates that we don't have necessary have the index for. You are told that the films have been deposited on the same substrate as Sub1. For the real part of PMMA-DR1 index, we will take an initial value of n=1.5.

Data:

	Si1	Si2	Si3
n	1,614	1,658	1,745
e (nm)	50,1	80,4	141
Ψ (°)	25,14	37,36	35,43
Δ (°)	81,52	62,90	300,58

-If we were to deposit the same dye on glass, what would be the effect that could perturb your measurement? What can we do to the back side of the glass substrate to avoid that?

-For a deposition of the same thickness (nominally) on glass and on silicon, would we effectively find the same thickness? If not, what could be the reason?

iii- Study of the homogeneity of a deposition

To test the homogeneity of a deposition, you have a sample of SiO_2 on Si (layer of SiO_2 obtained by thermal oxidation of silicon). Measure the different « measurable » optical parameters for oxide layer.

You observe different colours on the silicon oxide sample. Explain simply their origin. Try to link the difference in colour with a thickness difference. Conclude for the precision of the human eye to measure thicknesses… by interferometry.

Data:

Color	Thickness e (nm)
Green	65,5
Green/Blue	64,7
Blue	53,1
Purple	33,7

Clue: We will use the same type of configuration as for the Transmission lab work but we will look at the dephasing in reflection rather than in transmission.

v- Precision of the thickness measurement by ellipsometry

During this lab work, you shall have estimated the precision of the equipment, typically on the order of 1/10 of a nanometer. How can you simply explain such a precision?

Clue: we will use the expression of the dephasing introduce by the propagation of light in a thin film of thickness e and index n and consider that the precision on the measurement of this dephasing is 1/10 of a degree. Once again, we will use the same type of configuration as for the Transmission lab work but we will look at the dephasing in reflection rather than in transmission.